Name: ______________________

FUNCTIONS AND GRAPHS, (TRANSFORMATIONS OF GRAPHS)

Date: ________________

LO: I can describe and apply translations and reflections to the graphs of functions.

Graph transformations

A **translation** shifts a graph up, down, left or right. Adding a constant outside the function moves it **vertically**; adding inside moves it **horizontally** (in the opposite direction).

Reflections: $y = -f(x)$ reflects in the x-axis; $y = f(-x)$ reflects in the y-axis.

Transformation rules

🟢	$y = f(x) + 3$ shifts up 3	vertical translation
🟢	$y = f(x - 2)$ shifts right 2	horizontal translation
🟢	$y = -f(x)$ reflects in x-axis	reflection
🔴	$y = f(x + 2)$ shifts right 2	it shifts LEFT 2
🔴	$y = f(x) - 3$ shifts up 3	it shifts DOWN 3

Key idea

Changes outside f(x) affect the y-values (vertical). Changes inside f(x) affect the x-values (horizontal, opposite direction).

$$y = f(x - a) + b$$

Shift right a, then up b

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – vertical translation

The graph of $y = x^2$ is translated up 5 units. Write the new equation.

Start with $y = x^2$

Add 5 outside: $y = x^2 + 5$

Every point moves up 5

$\boxed{y = x^2 + 5}$

Adding a constant outside shifts the graph vertically.

Example 2 – horizontal translation

Describe the transformation from $y = x^2$ to $y = (x - 3)^2$.

Compare: x is replaced by $(x - 3)$

This is $f(x - 3)$ form

Translation 3 units to the right

Translation of 3 units to the right

Inside the bracket: subtract means move right.

Example 3 – reflection in x-axis

Sketch $y = -x^2$ given the graph of $y = x^2$.

$y = -f(x)$ reflects in the x-axis

Every y-value becomes its negative

U-shape becomes an inverted U-shape

Inverted parabola (reflected in x-axis)

All positive y-values become negative and vice versa.

Example 4 – combined transformation

Describe the transformation from $y = x^2$ to $y = (x + 1)^2 - 4$.

$(x + 1)$ means shift left 1

$- 4$ means shift down 4

Combined: translation $(-1, -4)$

Translation of (-1, -4): left 1 and down 4

Apply horizontal shift first, then vertical shift.

YOUR TURN – PRACTICE (deliberate practice)

1. Vertical translations

Write the equation after translating $y = x^2$ by the given amount.

a) Up 4

b) Down 7

c) Up 1

d) Down 10

2. Horizontal translations

Describe the transformation from $y = x^2$ to each new equation.

a) $y = (x - 4)^2$

b) $y = (x + 2)^2$

c) $y = (x - 1)^2$

d) $y = (x + 5)^2$

3. Reflections

Describe the transformation applied to $y = f(x)$.

a) $y = -f(x)$

b) $y = f(-x)$

c) $y = -x^3$

d) $y = (-x)^2$

4. Combined transformations

Describe the full transformation from $y = x^2$ to each equation.

a) $y = (x - 2)^2 + 3$

b) $y = (x + 4)^2 - 1$

c) $y = -(x - 1)^2$

d) $y = (x + 3)^2 + 5$

5. Mixed problems

Write the equation of $y = x^2$ after each transformation.

a) Reflect in x-axis then shift up 2 **b) Shift right 3 then down 1** **c) Shift left 2 then reflect in x-axis** **d) Shift up 6** **e) Shift right 5 then up 4**

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) $y = f(x) + 3$ translates the graph up 3 units ☐	**b) $y = f(x + 2)$ translates the graph right 2 units** ☐	**c) $y = -f(x)$ reflects the graph in the x-axis** ☐	**d) $y = f(x - 5)$ translates the graph left 5 units** ☐	**e) Vertical translations change the y-intercept** ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________	Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The curve $y = x^2$ is transformed to pass through the point **(3, 0)** as its minimum. Write down the equation of the new curve. Describe the single transformation.

Expression: ____________________________

Simplified: ____________________________

b) A function f(x) has a maximum at **(2, 5)**. State the coordinates of the maximum of: (i) $y = f(x) + 3$, (ii) $y = f(x - 1)$, (iii) $y = -f(x)$.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!